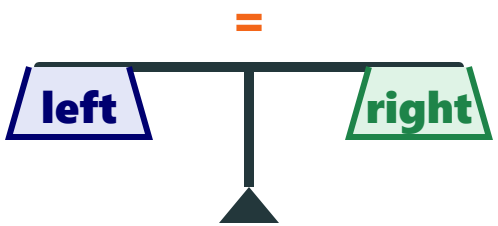
How to isolate a quantity in d = m / V



An equation is like a **balanced scale**: the two sides are equal.
If we do the **same operation on both sides** (multiply or divide by the same thing), the scale stays balanced. We use this to leave the quantity we want **alone**.

I want the mass, m

1. Write the formula. $$d = \frac{m}{V}$$
2. **V** is dividing **m**: multiply **both sides** by V. $$d \cdot V = \frac{m \cdot \cancel{V}}{\cancel{V}}$$
3. V/V = 1, so it cancels. Turn it round. $$\boxed{m = d \cdot V}$$

I want the volume, V

1. Start from m = d · V (steps on the left). $$m = d \cdot V$$
2. **d** is multiplying **V**: divide **both sides** by d. $$\frac{m}{d} = \frac{\cancel{d} \cdot V}{\cancel{d}}$$
3. d/d = 1, so it cancels. Turn it round. $$\boxed{V = \frac{m}{d}}$$

Shortcut: what is **dividing** on one side goes to the other side **multiplying**, and what is **multiplying** goes to the other side **dividing**.